

Pilot stored elements kept across sessions

Optional flypad setting that keeps the MFD pilot-stored waypoints, nav aids, routes and runways

Branch	feature/a380/mfd/pilot-stored-elements-persistence
Aircraft	A380X (setting shared)
Commits	7636abc1c feat(a380x/mfd): keep the pilot stored elements across sim sessions (flypad setting)
Base	pr/10-a380x-mfd-fcom-pages (first built on the old fork master 280800cf7)
Status	Merged in develop. Harness-tested over two sessions.

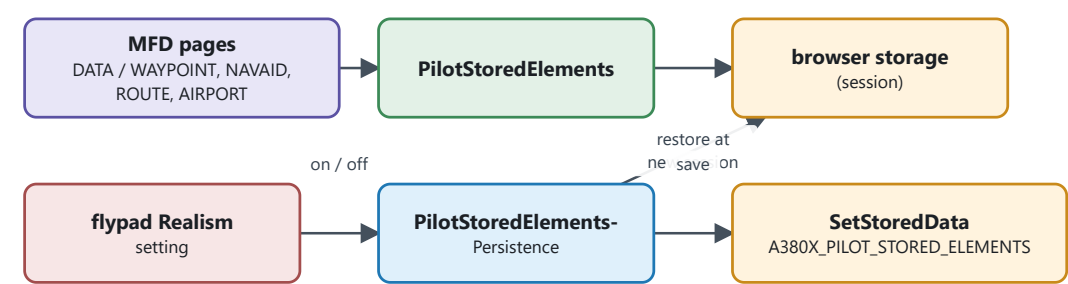
Purpose

Stored waypoints, nav aids, routes and runways disappeared when the sim closed. The FCOM says "All the pilot-stored elements are deleted when all FMCs are shut down" (DSC-22-FMS), so that stays the default; the user chose an optional flypad setting to keep them.

Specification

- Realism page setting **Keep pilot stored elements** (off by default = FCOM behaviour, marked unrealistic).
- While on: the browser-storage keys of the stored elements (`A32NX.StoredWaypoints` , `A380X.PilotStored{Nav aids, Routes, Runways}`) are saved to the persistent storage (`SetStoredData` `A380X_PILOT_STORED_ELEMENTS`).
- At a new session (sentinel key `A380X.PilotStoredElementsSession`) they are restored before the FMS loads them.

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture

Rules

Rule	Source	Implementation
Pilot-stored elements deleted when all FMCs are shut down	A380 FCOM DSC-22-FMS	default behaviour kept (setting off)

PR description (from the PR draft)

From `pr-drafts/13-pilot-stored-elements-persistence.md`, section "Summary of Changes" onwards.

Summary of Changes

The A380 FCOM says "All the pilot-stored elements are deleted when all FMCs are shut down" (DSC-22-FMS); that stays the default. A new flypad realism setting, **Keep pilot stored elements** (off by default, marked unrealistic), saves the stored waypoints, nav aids, routes and runways in the persistent storage (`SetStoredData A380X_PILOT_STORED_ELEMENTS`) and restores them at the next session, before the FMS loads them (a session sentinel tells a new session from a reload).

Also in this PR:

- The OIT flyPad (OITLegacy) gets the new realism option too, with the A380X EFB value; without it the `FBW_TYPECHECK` build fails.
- 17 unit tests (`PilotStoredElements.spec.ts`): the FCOM database limits (20 NAVAIDs and 10 runways, the first created one deleted when full; 5 routes, then PILOT RTEs LIST FULL), replacement instead of duplicates, city pair routes, deletions, the NAVAID class fields, unreadable storage, and the setting across sessions (kept or deleted, written only on a change, not restored twice in a session, the kept copy dropped when the setting is turned off).
- Test setup fix (`fbw-common/src/jest/setupJestMock.ts`, test only): Node 22+ defines its own `localStorage` global, undefined without `--localstorage-file`, and vitest keeps it instead of the jsdom one. The setup now uses the jsdom storage, or an in-memory one, when `localStorage` is undefined. The new tests need it, and it also makes the existing `WaypointEntryUtils.spec.ts` and `SimBriefUplinkAdapter.spec.ts` load again on Node 22+ (they failed to load in the dev container, Node 26).

Cockpit API Changes

None. New stored data key `A380X_PILOT_STORED_ELEMENTS`, new setting `KEEP_PILOT_STORED_ELEMENTS`.

Screenshots (if necessary)

TO ADD: the Realism setting; DATA / WAYPOINT after a sim restart with the setting on.

References

- A380 FCOM DSC-22-FMS (pilot-stored elements deleted at FMC shutdown).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Setting off: store a waypoint, restart the sim: it is gone (FCOM).
2. Setting on: store waypoints, nav aids, a route and a runway, restart the sim: they are back.
3. OIT flyPad: Settings > Realism shows the option.
4. `npx vitest run fbw-a380x/src/systems/instruments/src/MFD/FMC/PilotStoredElements.spec.ts` (17 tests), and the full `npm test` on Node 22+ (the two existing specs above load again).